



CSCE 633-700: Machine Learning

Support Vector Machines

Overview

- Lagrange Optimization
- Support Vector Machines
 - Linearly separable case
 - Non-separable case
 - Hinge Loss
 - Multi-class SVMs

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Upper bound estimate of expected number of errors

- The number of support vectors is an upper bound estimate of the expected number of errors (Vapnik, 1995)

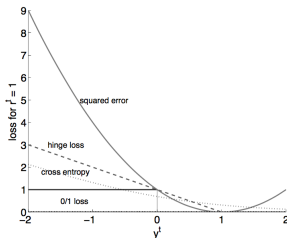
$$\mathbb{E}_N[P(error)] \leq \frac{\mathbb{E}_N[\# \text{support vectors}]}{N}$$

- The error rate depends on the number of support vectors and not the feature dimensionality

Support Vector Machines: Hinge Loss

- Decision rule: $f(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + w_0)$
 - $f(\mathbf{x}) = 1$, if $\mathbf{w}^T \mathbf{x} + w_0 > 0$
 - $f(\mathbf{x}) = -1$, if $\mathbf{w}^T \mathbf{x} + w_0 < 0$
- If $f(\mathbf{x})$ is the output and y_n the actual label

$$l^{\text{hinge}}(f(\mathbf{x}), y) = \begin{cases} 0 & \text{if } y(\mathbf{w}^T \mathbf{x} + w_0) \geq 1 \\ 1 - y(\mathbf{w}^T \mathbf{x} + w_0) & \text{otherwise} \end{cases}$$



- Hinge loss penalizes incorrectly classified samples more than 0/1 loss and cross-entropy loss
- Hinge loss penalizes correctly classified samples as well